Synthetic sample document - fictional data for demos and evaluation only.

STANDARD OPERATING PROCEDURE: HEAT EXCHANGER E-505 CLEANING
Document ID: SOP-HX-505
Equipment: Shell-and-tube cooling water heat exchanger E-505

1. Performance Monitoring:
- Record inlet and outlet temperatures on both sides weekly.
- Schedule cleaning when the overall heat transfer coefficient falls 25 percent below its clean value, or when tube-side pressure drop exceeds 1.8 bar.
- Trend the results so fouling is caught before production is affected.

2. Cleaning Method:
- Isolate and drain the exchanger following SOP-LOTO-001.
- Remove the channel head and inspect the tube sheet.
- Hydroblast the tubes at 700 bar with a rotating nozzle.
- For hard scale, circulate 5 percent citric acid at 60 degC for 4 hours, then neutralize to pH 6 to 8.
- Never use hydrochloric acid on stainless steel tubes.

3. Reassembly and Testing:
- Replace all gaskets with new spiral wound gaskets.
- Torque the channel head bolts in a cross pattern to 450 Nm.
- Hydrostatic test at 1.5 times design pressure (15 bar) for 30 minutes with no visible leaks.
- Return the exchanger to service and record the new clean-condition temperatures.